MODULE 08: INDUCTION & INDUCTANCE

(Dr. Rubaiyet I. Haque)

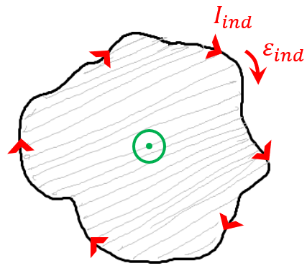
LECTURE 18

OUTLINE:
- Induced electric field
- Inductors and inductance
- Self-induction

INDUCED ELECTRIC FIELD

An induced electric field is an electric field that is created because of a changing magnetic field. When there is a change in the magnetic field within a closed loop or circuit, an induced electric field is generated within the circuit.

Let us consider, a copper ring of radius r is placed in a uniform external magnetic field that is coming out of plan, as shown in Figure.



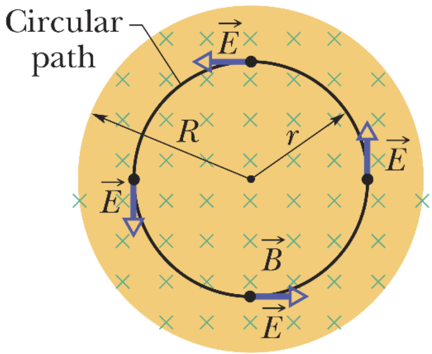
- If the magnetic field strength increases at a steady rate, due to increasing the current flow in the windings of the electromagnet that produces the field.
    - Magnetic flux through the ring is then changed at a steady rate.
    - This leads to the formation of an induced emf and thus an induced current appears in the ring.
    - As per Lenz's law, direction of induced current is clockwise.
- Therefore, an electric field must be present along the ring, because an electric field is needed to do the work of moving the conduction electrons.
    - Moreover, the electric field must have been produced by the changing magnetic flux. This is known as ***induced electric field***.
    - A changing magnetic field produces an electric field, even if the changing magnetic field were in a vacuum (even if there is no conductive copper ring).

The magnitude and direction of the induced electric field depend on the rate of change of the magnetic field and the geometry of the conducting loop. The induced electric field follows certain principles, such as:

- *Faraday's Law*: The magnitude of the induced electric field is directly proportional to the rate of change of the magnetic field. A larger rate of change of the magnetic field induces a stronger electric field.
- *Lenz's Law*: The induced electric field creates an electric current that opposes the change in the magnetic field causing it. It acts in a way that tends to preserve the existing magnetic field.

REFORMULATION OF FARADAY'S LAW

Let us consider a particle of charge q_0 is moving around the circular path (Fig.). Thus, the work (W) done on the charged particle in one revolution by the induced electric field is,



$$W = q_0 \varepsilon_{ind} \qquad [\because W = q\Delta V]$$

where ε_{ind} is the induced emf, known as the work-done per unit charge in moving the positive test charge q_0 around the path.

Again, the work can also be expressed as,

$$W = \oint \vec{F}.d\vec{s} = \oint q_0\vec{E}.d\vec{s} = q_0 \oint \vec{E}.d\vec{s} \qquad [\because \vec{F} = q_0\vec{E}]$$

Therefore,

$$W = q_0\varepsilon_{ind} = q_0 \oint \vec{E}.d\vec{s} \qquad \Rightarrow \varepsilon_{ind} = \oint \vec{E}.d\vec{s}$$

Therefore, an induced emf is the sum of quantities $\vec{E}.d\vec{s}$ around a closed path, where $\vec{E}$ is the electric field induced by a changing magnetic flux and $d\vec{s}$ is a differential length vector along the path.

Therefore, Faraday's law can be written as,

$$\varepsilon_{ind} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{E}.d\vec{s} = -\frac{d\Phi_B}{dt} = -\frac{\partial}{\partial t}\oiint \vec{B}.d\vec{A}$$

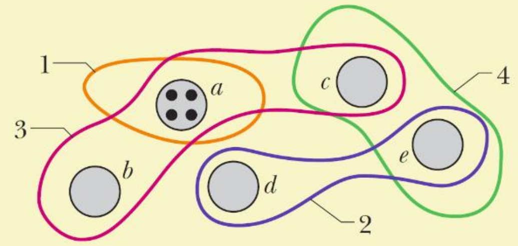
For a uniform electric field (E), We get,

$$\varepsilon_{ind} = \oint \vec{E}.d\vec{s} = \oint E\, ds = E\oint ds = E(2\pi r) = 2\pi r E$$

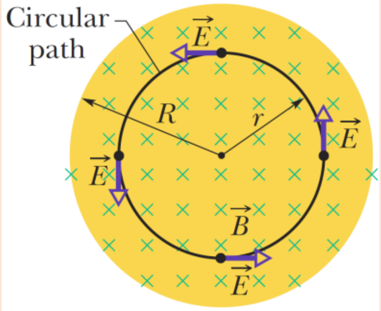
where $s = 2\pi r$ is the circumference of the circular path over which the force acts.

CHECKPOINT: The figure shows five lettered regions in which a uniform magnetic field extends either directly out of the page or into the page, with the direction indicated only for region *a*. The field is increasing in magnitude at the same steady rate in all five regions; the regions are identical in area. Also shown are four numbered paths along which have the magnitudes given below in terms of a quantity "mag." Determine whether the magnetic field is directed into or out of the page for regions *b* through *e*.

Path	1	2	3	4
$\oint \vec{E}\cdot d\vec{s}$	mag	2(mag)	3(mag)	0



ANSWER: $b \rightarrow out;\ c \rightarrow out;\ d \rightarrow into;\ e \rightarrow into$



PROBLEM: In Figure, take $R = 8.5\ cm$ and $\frac{dB}{dt} = 0.13\ T/s$.

(a) Find an expression for the magnitude E of the induced electric field at points within the magnetic field, at radius r from the center of the magnetic field. Evaluate the expression for $r = 5.2\ cm$.

(*b*) Find an expression for the magnitude E of the induced electric field at points that are outside the magnetic field, at radius r from the center of the magnetic field. Evaluate the expression for $r = 12.5\ cm$.

ANSWER: (a) From Faraday's law we know,

$$\oint \vec{E}.d\vec{s} = -\frac{d\Phi_B}{dt} \ldots \ldots \ldots (1)$$

We know,

$$\oint \vec{E}.d\vec{s} = \oint E\ ds = E \oint ds = E(2\pi r) \ldots \ldots \ldots (2)$$

(The integral $\oint ds$ is the circumference $2\pi r$ of the circular path.)

Because $\vec{B}$ is uniform over the area A encircled by the path of integration and is directed perpendicular to that area, the magnetic flux is given by,

$$\Phi_B = BA = B(\pi r^2)$$

Substituting this and Eq. (2) into Eq. (1) and dropping the minus sign, we find that,

$$E(2\pi r) = (\pi r^2)\frac{dB}{dt}$$

$$\text{Or, } E = \frac{r}{2}\frac{dB}{dt} \ldots \ldots \ldots (3)$$

Equation (3) gives the magnitude of the electric field at any point for which $r \leq R$ (that is, within the magnetic field). Substituting given values yields, for the magnitude of $\vec{E}$ at $r = 5.2\ cm$,

$$E = \frac{(5.2 \times 10^{-2}\ m)}{2}(0.13\ T/s) = 0.0034\frac{V}{m} = 3.4\ mV/m$$

(b) We know,

$$\Phi_B = BA = B(\pi R^2)$$

Substituting this and Eq. (2) into Eq. (1) (without the minus sign) and solving for E yield

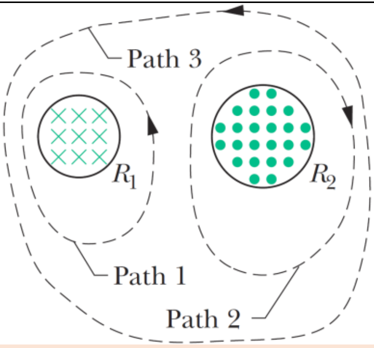
$$E = \frac{R^2}{2r}\frac{dB}{dt} \ldots \ldots \ldots (4)$$

Because E is not zero here, we know that an electric field is induced even at points that are outside the changing magnetic field, an important result that makes transformers possible.

With the given data, Equation (4) yields the magnitude of $\vec{E}$ at $r = 12.5\ cm$,

$$E = \frac{(8.5 \times 10^{-2}\ m)^2}{2(12.5 \times 10^{-2}\ m)}(0.13\ T/s) = 3.8 \times 10^{-3}\frac{V}{m} = 3.8\ mV/m$$

PROBLEM 30-36: Figure shows two circular regions R_1 and R_2 with radii $r_1 = 20.0\ cm$ and $r_2 = 30.0\ cm$. In R_1 there is a uniform magnetic field of magnitude $B_1 = 50.0\ mT$ directed into the page, and in R_2 there is a uniform magnetic field of magnitude $B_2 = 75.0\ mT$ directed out of the page (ignore fringing). Both fields are decreasing at the rate of $8.50\ mT/s$. Calculate $\oint \vec{E}.d\vec{s}$ for (a) path 1, (b) path 2, and (c) path 3.



PROBLEM 30-39: The magnetic field of a cylindrical magnet that has a pole-face diameter of $3.3\ cm$ can be varied sinusoidally between $29.6\ T$ and $30.0\ T$ at a frequency of $15\ Hz$. (The current in a wire wrapped around a permanent magnet is varied to give this variation in the net field.) At a radial distance of 1.6 cm, what is the amplitude of the electric field induced by the variation?

INDUCTORS AND INDUCTANCE

Inductor:

An inductor is a passive electronic component used in electrical circuits to store energy (*magnetic energy*) in the form of a magnetic field when an electric current flows through it.

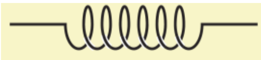


Fig: *Inductor*

- It consists of a coil or winding of wire typically wrapped around a core.
  - The wire used in an inductor is usually made of a conductive material like copper.
  - Core could be made of various materials, such as, iron, ferrite, or air.
- Inductors are widely used in electrical and electronic circuits for various purposes, including energy storage, filtering, signal processing, and impedance matching.
- When electric current flowing through an inductor change, it induces an EMF (back emf) that opposes the change in current. If L is the inductance, di/dt refers to the rate of change in current over time. The potential difference across the inductor V_L is described by the expression,

$$V_L = -L\frac{di}{dt}$$

Inductance:

Inductance is a measure of the ability of an inductor to store energy in the form of a magnetic field when an electric current passes through it.

- In other words, inductance is the ratio of the magnetic flux linked by the inductor's coil to the current flowing through the coil.
- Inductance is denoted by the letter L.
- The SI unit of inductance is the tesla - square meter per ampere ($T.m^2/A$), which is also known as the henry (H). $1\ henry = 1\ H = 1\ T.m^2/A$

If a current i is established through each of the N windings of an inductor, a magnetic flux Φ_B links those windings. Mathematically, the inductance L can be expressed as,

$$L = \frac{N\Phi_B}{i}$$

Where N is the number of turns of the coil. The windings of the inductor are said to be *linked* by the shared flux, and the product $N\Phi_B$ is called the *magnetic flux linkage.* The inductance L is thus a measure of the flux linkage produced by the inductor per unit of current.

Inductance of a Solenoid:

A solenoid is a type of inductor formed by a tightly wound coil of wire. The inductance of a solenoid is primarily determined by its physical characteristics, such as the number of turns, the length of the coil, the cross-sectional area, and the presence of a magnetic core.

Let us consider, a long solenoid having cross-sectional area A. Considering the length l near the middle of this solenoid. The flux linkage there is,

$$N\Phi_B = (nl)(BA)$$

Where n is the number of turns per unit length of the solenoid and B is the magnitude of the magnetic field within the solenoid. We know, the magnitude B of a solenoid is given by,

$$B = \mu_0 in$$

Therefore, inductance can be expressed as,

$$L = \frac{N\Phi_B}{i} = \frac{(nl)(BA)}{i} = \frac{(nl)(\mu_0 in)(A)}{i} = \mu_0 n^2 lA \qquad \left[\because L = \frac{N\Phi_B}{i}\right]$$

Thus, the inductance per unit length near the center of a long solenoid is,

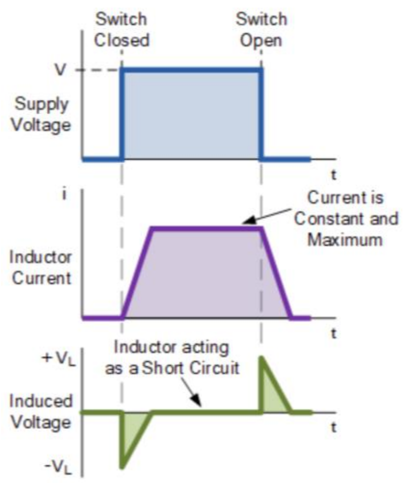
$$\frac{L}{l} = \mu_0 n^2 A$$

Flux linkage: Flux linkage is a measure to quantify the amount of magnetic flux linking or passing through a circuit or component. It is a measure of how many magnetic field lines are linked with a given circuit or coil.

SELF-INDUCTION

Self-induction is a phenomenon in electromagnetism that describes the tendency of a current carrying inductor or conducting wire loop to generate an electromotive force (EMF) or voltage within the same coil or circuit when the current passing through it changes.

- When the current through an inductor increases or decreases, the inductor generates a voltage in such a way that it opposes the change in current.
- Thus, self-inductance can be thought of as the ***inductor's resistance*** to changes in current.
- ***Self-inductance allows inductors to store energy*** in the form of a magnetic field.



The current flowing through an inductor creates a magnetic field around the inductor. The change in current through the inductor results the change in magnetic field through the induces. This leads to generation of induced emf (back emf) in the same inductor, opposing the change in current flow through the inductor itself. The ***induced emf*** is directly proportional to the rate of change of current and is given by the equation,

$$\varepsilon_L = -L\frac{di}{dt}$$

Since, $L = \frac{N\Phi_B}{i}$ $\Rightarrow Li = N\Phi_B$ $\Rightarrow L\frac{di}{dt} = N\frac{d\Phi_B}{dt}$

Therefore, $\varepsilon_{ind} = -L\frac{di}{dt}$ $\left[\because \varepsilon_{ind} = -N\frac{d\Phi_B}{dt}\right]$

where ε_L is the induced electromotive force, L is the self-inductance of the circuit or coil, di/dt is the rate of change of current with respect to time.

- The self-inductance, represented by the symbol L, is a measure of the ability of a circuit or coil to generate an induced emf due to its own changing current.
- It depends on factors such as the number of turns in the coil, the geometry of the coil, and the presence of any magnetic core material.

The self-inductance of a circuit can be determined experimentally or calculated using the formula:

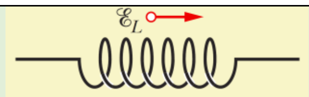
$$L = \frac{\mu_0\mu_r N^2 A}{l} \qquad [since, \quad L = \mu_0 n^2 lA \quad \& \quad N = nl]$$

where μ_0 is the permeability of free space (approximately $4\pi \times 10^{-7}\ H/m$), μ_r is the relative permeability of the core material (for air, $\mu_r = 1$), N is the number of turns, A is the cross-sectional area of the coil, and l is the length of the coil.

Self-inductance has several important implications:

- <u>*Inductive Kickback*</u>: When the current through an inductor is suddenly interrupted, the self-induction creates a counter emf that can result in a high voltage spike. This effect is commonly referred to as inductive kickback or back emf and can be potentially damaging to electronic components if not properly managed.
- <u>*Energy Storage*</u>: Self-inductance allows the storage of energy in the magnetic field surrounding the coil. This energy can be released when the current changes, contributing to the operation of devices such as relays, transformers, and inductors in power supplies.
- <u>*Time Delays*</u>: Self-inductance causes a time delay in the response of the current to changes in the applied voltage. This delay can have implications in circuits that require precise timing or control.

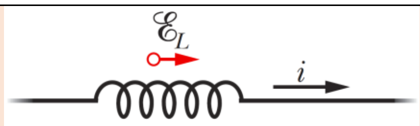
CHECKPOINT: The figure shows an emf ε_L induced in a coil. Which of the following can describe the current through the coil: (a) constant and rightward, (b) constant and leftward, (c) increasing and rightward, (d) decreasing and rightward, (e) increasing and leftward, (f) decreasing and leftward?



ANSWER: d and e

PROBLEM 30-44: A 12 H inductor carries a current of 2.0 A. At what rate must the current be changed to produce a 60 V emf in the inductor?

PROBLEM 30-45: At a given instant the current and self-induced emf in an inductor are directed as indicated in figure. (a) Is the current increasing or decreasing? (b) The induced emf is 17 V, and the rate of change of the current is 25 kA/s; find the inductance.



PROBLEM 30-46: The current i through a 4.6 H inductor varies with time t as shown by the graph of figure, where the vertical axis scale is set by $i_s = 8.0\ A$ and the horizontal axis scale is set by $t_s = 6.0\ ms$. The inductor has a resistance of 12 Ω. Find the magnitude of the induced emf ε during time intervals (a) 0 to 2 ms, (b) 2 ms to 5 ms, and (c) 5 ms to 6 ms. (Ignore the behavior at the ends of the intervals.)